Mobile satellite communications proposals

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The next generation of satellite networks, aimed at a hand-held voice and data terminal market similar to that of GSM, is currently under development for launch of services within the next five years. This paper reviews the leading proposals.

1. Introduction

The past five years has seen a number of proposals for serving hand-held voice and data terminals directly by satellite. While there has been considerable hype over individual system capabilities and schedules, over one third of the investment needed to launch four competing global mobile satellite systems has been secured. With prototype spacecraft likely to be launched during 1996, initial commercial services can be expected to be in operation by 1998—2001.

Four major players have emerged in the hand-held market — ICO (a derivative of Inmarsat), Motorola's Iridium, Loral/Qualcomm's Globalstar, and Odyssey from TRW. Each system has its own characteristic system architecture, orbits, technologies and approach to local service provision.

The capacity of these satellite systems is very small compared with terrestrial cellular and poses more opportunity than threat to national cellular network operators and service providers. Coverage of satellite systems is limited in urban and indoor environments because of tight power budgets and terrestrial clutter shadowing the radio path; this narrows target applications to certain niche markets such as international business travellers, cellular extension into remote locations and areas where communications infrastructure has yet to be developed. In each case, global operation is the prime requirement.

2. The interest in hand-held satellite voice communications

Research towards hand-held voice and data terminal communications via satellite has been progr essed for many years by organisations such as the European Space Agency (ESA) with its Archimedes and Artemis programmes, Japan's National Space Development Agency (NASDA) with its various engineering test satellite programmes and the US National Aeronautics and Space Agency (NASA) with its advanced communications technology satellite programme. More recently Motorola precipitated the debate of how to proceed with the launch of commercially viable satellite personal communications networks by announcing its plans for a system called Iridium in June 1990. This announcement concentrated the minds of the other potential key players in the satellite and mobile communications arenas who have responded variously with comments on the Iridium proposals and with proposals for their own, alternative systems.

At present the most visible activity is in the USA where the Federal Communications Commission (FCC) has reviewed its options to determine which proposals to authorise. Inmarsat, the organisation responsible for the majority of mobile satellite communications outside the USA has responded with its own proposal for a future system — ICO.

3. US proposals for the RDSS bands

In the USA, hand-held mobile satellite voice communications is historically linked with the provision of position fixing Radio Determination Satellite Service (RDSS). Some years ago the FCC licensed a number of companies to provide RDSS and messaging facilities within a set of frequency bands assigned specifically to these services. Only one of these companies, Geostar, actually commenced service. Geostar suffered from a lack of uptake for its limited services and rapidly ran into financial trouble. It subsequently applied to the FCC for authority to allow substantial alterations to its system, modifications that were tantamount to a new system. In May 1991 the FCC rejected Geostar's revised proposals and Geostar discontinued service on 13 May 1991. This left the bands clear apart from the low-power spread spectrum signals of the Soviet Glonass positioning system and some experimental radio-astronomy use.

Meanwhile, in November 1990, the FCC's September 1990 Public Notice requesting comments on Geostar's new proposals drew an application from Ellipsat promising voice communications as well as RDSS and messaging facilities. This was closely followed in December 1990 by the eagerly awaited and even more ambitious Iridium application from Motorola which had been originally announced in June 1990. A number of other companies have since filed applications for their own systems and made comments on the proposals already before the FCC in response to the Motorola and Ellipsat petitions being put out for comments on 1 April 1991. The majority of these applications called for combined RDSS and voice communications to be allowed in the RDSS bands.

The deadline for further proposals and comments on Motorola's and Ellipsat's proposals passed and on 31 January 1995 the FCC authorised only three — Iridium, Globalstar and Odyssey. These systems all share the frequency bands designated for spread spectrum RDSS use in the USA in various ways and offer RDSS as an additional service to personal communications to qualify for use of these frequencies. The RDSS frequencies were:

L-band	1610.0 — 1626.5 MHz	mobile uplink,
S-band	2483.5 — 2500.0 MHz	mobile downlink,
C-band	6525.0 — 6541.5 MHz	FES uplink,
	5199.5 — 5216.0 MHz	FES downlink.

All the systems are designed to be used with mobile terminals of approximately the same size and weight as current terrestrial cellular telephones. Most of the consortia proposing these services intend to lease capacity on their systems to non-common carriers for resale to customers rather than acting as sole vendors of the services themselves. Cellular service providers in the USA and elsewhere are looking to these systems as a way to extend their coverage for roaming customers.

4. Iridium

Motorola Satellite Communications Inc of Chandler, Arizona, was the first company to propose a commercial voice carrying satellite network, with a system called Iridium [1]. They have been successful in attracting investment from a variety of other communications companies, from equipment manufacturers and satellite launchers to telecommunications network operators, to finance jointly the provision of the $3.7 billion satellite infrastructure. A total of US$1.6 billion had been invested by companies from all over the world including US Sprint, DDI of Japan, STET (Italy's PTT) and a host of equipment manufacturers and satellite launchers such as Lockheed, Raytheon, Great Wall Industries, Krunichev Enterprise, Sony, Mitsubishi, Mitsui and Koyocera. Lockheed Corp Missiles & Space Co has been selected to build the spacecraft.

Iridium is designed to offer voice services to US$3000 hand-held dual-mode Iridium/cellular terminals, the cellular part of the phone being compatible with a local terrestrial cellular standard such as GSM, PDC, D-AMPS or CDMA. Within compatible cellular coverage the handset will act like a terrestrial cellular telephone, switching to Iridium only where there is no compatible cellular coverage. On the Iridium network, call charges can be expected to be a flat US$3/minute for a call to anywhere in the world. Iridium's business case predicts 6 million Iridium terminals to be in use by 2001, 1.3 million of which will be in the USA. Iridium will also offer a position determination service, facsimile, paging, two-way messaging and duplex 2.4 kbit/s data bearers to terminals that are suitably equipped. A paging-only receiver or a position determination unit is expected to cost about US$300.

The Iridium constellation consists of 66 satellites in six different 780 km altitude circular near-polar orbital planes, each plane containing eleven equi-spaced satellites, as depicted in Fig 1. This constellation provides continuous

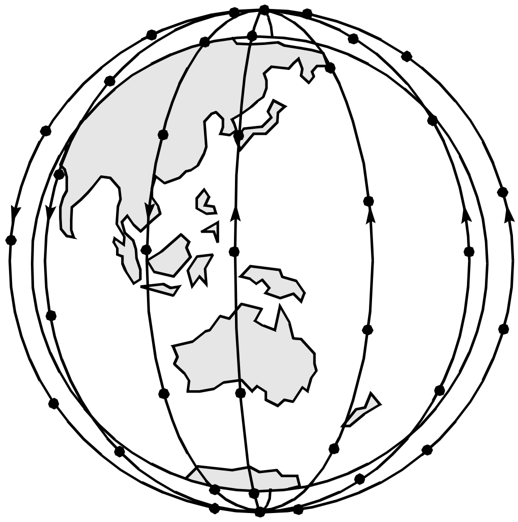


Fig 1 The Iridium constellation.

global coverage. Each satellite covers a circular footprint area of diameter 4600 km with 48 circular beams formed by three multiple-beam phased-array antennas — this coverage is just sufficient for complete coverage at the equator. The beams are arranged in a regular hexagonal pattern, shown in Fig 2, which moves across the Earth's surface with the satellite. Away from the equator, as the orbit tracks converge, beams are switched off to prevent beams overlapping too much which would either cause interference or reduce the opportunity for frequency reuse. Iridium does not use satellite diversity — communication is with only one satellite at a time and a 16 dB fading and shadowing margin allows the radio signal to penetrate radio path obstructions. Iridium reuses its time division multiple access (TDMA) channels in a seven-cell frequency reuse pattern that is geographically fixed on the Earth's surface;

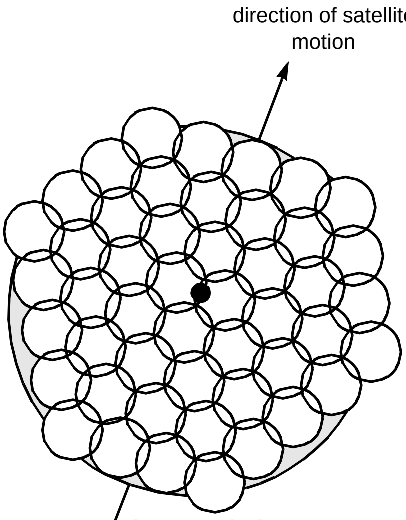


Fig 2   Iridium's regular beam pattern.

the satellite beams move over the cell pattern, adopting the cell frequencies as they move.

Mobile uplinks and downlinks are time-domain duplexed within the same frequency band, namely the L-band RDSS allocation. Uniquely among the FCC applicants to use this band, Iridium has an FCC waiver of rules to allow it not to use spread-spectrum in the RDSS band but to use TDMA instead. The mobile terminals transmit up to 3.7 W in bursts of 8.28 ms duration in every 90 ms time frame, averaging out to 0.34 W over time. Modulation is quadrature phase shift keying (QPSK) from 50 kbit/s modulators. Each TDMA frame has four transmit and four receive time-slots and a number of TDMA carriers are spaced at 41.67 kHz intervals throughout the frequency bands. Motorola plans to switch and route these circuits on board the satellites, involving on-board processing on a scale not previously seen; the most advanced satellite switching used commercially to date is satellite switched TDMA used on Italsat and Intelsat-VI satellites, operating at radio frequencies, rather than baseband as Motorola plans. With 4.8 kbit/s voice coding, rate $^{3}/_{4}$ convolutional FEC coding, control overheads and as many as 1100 users per satellite, satellite switching requirements are substantial.

Because of the low orbital altitude of the satellites and their small coverage footprints, satellites must communicate to relay information from satellites over the oceans to land-based fixed Earth stations (FESs). This also allows an FES close to the call's destination to be used, avoiding the use of terrestrial international networks for long-distance routeing. Inter-satellite radio links at 22 GHz and 33 GHz will be used to network the satellites together. Each satellite uses four 25 Mbit/s QPSK modems to transmit and receive data from four satellites adjacent to it (two in its own orbit and one in each of the adjacent orbital planes). The antennas for inter-satellite links are wave-guide slots with mechanically steered reflectors.

FES links will be in the S-band RDSS frequency band — Iridium does not use the RDSS C-band frequencies. FES links use two 6.25 Mbit/s QPSK modems to transmit and receive backhaul links. As half rate coding is used for FES and inter-satellite links, on-board processing must include error correction and re-coding as well as switching. With the 1100 mobile users and up to 5000 additional calls in transit through a satellite to FESs, the on-board processing required to support the dynamic F/TDMA switching is of the order of 100 Mbit/s throughput on each satellite.

5. Globalstar

Loral Cellular Systems Corp, a joint venture between Loral Aerospace Corp of Newport Beach, California and Qualcomm Inc of San Diego have proposed the Globalstar system [2]. Loral Aerospace (formerly Ford Aerospace) will build the satellites and Qualcomm provide the expertise in CDMA cellular systems. Globalstar have also been successful in attracting investment for their system from international investors. US$784 million has been secured from investors including equipment manufacturers Alcatel, Deutsch Aerospace, Hyundai and Alenia Spazio as well as cellular operators AirTouch (a Pacific Telesis offshoot), Dacom (Korea) and Vodafone. Total investment is predicted to be US$2.6 billion to complete the satellite network and FES infrastructure.

Globalstar service will always be sold through local operators and is pitched at US$0.65 per minute access charge plus international and trunk terrestrial routeing charges for the call to reach its destination. A dual-mode handset is anticipated to cost as little as US$500.

The Globalstar constellation consists of 48 satellites in circular orbits in eight planes at 1414 km altitude. The orbits are inclined at 52° to the equator with six satellites in each orbital plane. This provides full coverage of the globe up to ± 80° latitude at all times from two satellites at the same time, as shown by Fig 3. This dual-satellite coverage will be

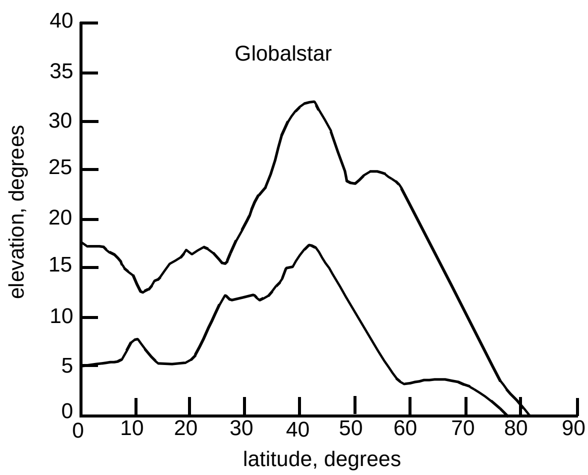


Fig 3   Minimum elevation angles for two Globalstar satellites.

used for satellite diversity to avoid shadowing of the radio link by terrestrial clutter. Use of CDMA makes satellite diversity simple, using CDMA's soft handover technique. Each satellite's coverage is split into 16 beams as shown in Fig 4, with the same frequencies reused in every beam. The elongated beam pattern minimises handover between beams as the satellite passes over users.

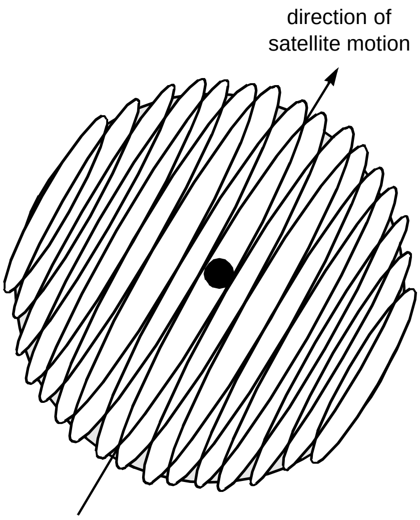


Fig 4 Globalstar's elongated spot beams.

Mobile uplink and downlink are frequency division duplexed into different bands. Uplink uses RDSS L-band allocation and downlink uses RDSS S-band allocation. To accommodate the extra traffic in the FES links the C-band requirement is extended to include some of the FSS spectrum next to the RDSS C-band spectrum.

The mobile up and downlink bands are divided into 13 × 1.25 MHz CDMA sub-bands, all used in all cells. Globalstar uses variable rate voice coding at rates between 1.2 and 9.6 kbit/s, averaging 4.8 kbit/s over time. FEC coding is applied at rate $^1/_3$ for the uplink and rate $^1/_2$ for the downlink. These signals are direct-sequence spread to the 1.25 MHz sub-band bandwidth. Transmit power is up to 2 W but only while the user is speaking. Modulation is QPSK.

Globalstar transponders are transparent 'bent-pipe' repeaters routeing all traffic directly between the mobile terminals and FESs on the ground where control is centralised. FESs act independently from each other, giving FES operators full control over call routeing. The 7600 km diameter coverage footprint of each satellite is large enough to ensure that intended traffic can always be routed to an FES within the satellite footprint, so inter-satellite links are not required.

It should be noted that Qualcomm have been successfully using their 'Qualcomm Automatic Satellite Position Reporting' techniques in operating their OmniTRACS Geostationary RDSS and messaging service in the USA since May 1990 and in Europe (where it is known as EutelTRACS and operated by Eutelsat) since January 1990. It is on this system that the RDSS services of Globalstar will be based. Furthermore, Qualcomm were designers of IS95, the US terrestrial cellular network standard using the same 1.25 MHz chip-rate direct sequence CDMA technique as planned for Globalstar. Qualcomm wishes to exploit this common ground between the two systems with dual-mode mobile terminals capable of communicating both through satellite and through the terrestrial network. The changes required to communicate in either manner may be as simple as frequency translation of the spread spectrum signal. The terrestrial CDMA system has already been deployed in commercial service in some US PCS networks.

6. Odyssey

TRW Inc of Cleveland submitted an application to the FCC on 31 May 1991 to launch the Odyssey system [3]. TRW is seeking partners to own and operate Odyssey but does not appear to be as active as Iridium, Globalstar or ICO in finding partners. Odyssey's system is the lowest cost of the authorised FCC proposals at US$2.3 billion for the satellites and FES infrastructure.

The constellation, shown in Fig 5, consists of twelve satellites, four in each of three circular orbits at 10 354 km altitude and 55° inclination. As Fig 6 shows, this provides the opportunity for full global coverage with two or three satellites visible from any location on Earth at any time. However, the satellite antennas will point at and track their given service areas to ensure that extra capacity is available over the US, Europe and the Pacific Rim while still maintaining coverage of the industrial world. Odyssey does not plan to use satellite diversity but instead exploit the high elevation angle of its satellites to avoid shadowing from outdoor terrestrial clutter. Each satellite has antennas to create 37 beams for mobile uplinks and downlinks, each of which can be steered to track a geographic area.

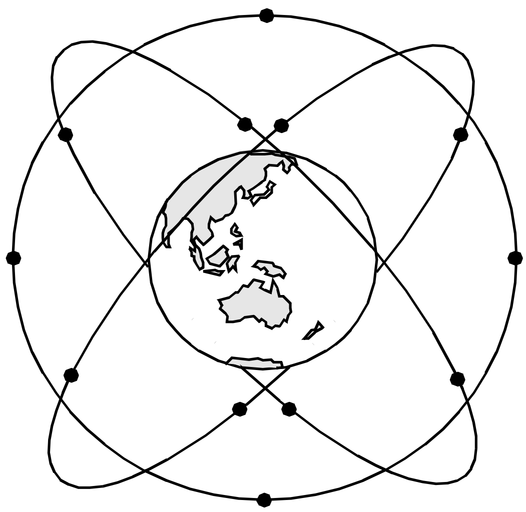


Fig 5 Odyssey constellation.

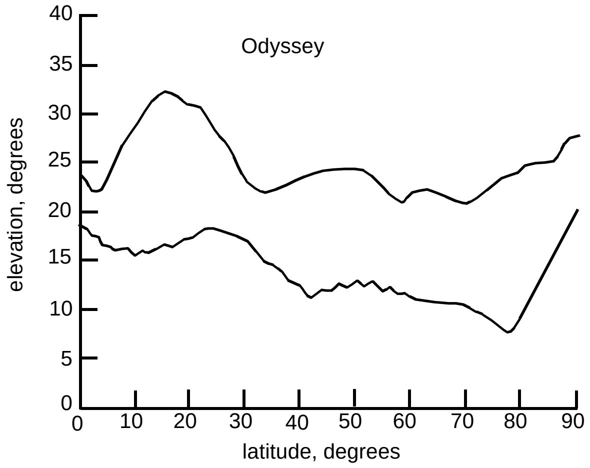


Fig 6 Minimum elevation angles for two Odyssey satellites.

Mobile uplinks will be contained in the RDSS L-band allocation and downlinks in RDSS S-band. Each of these is divided into 3 × 4.833 MHz sub-bands. FES links are in Ka-band at 30.0 GHz for the uplink and 20 GHz for the downlink. The RDSS C-band frequencies are not used.

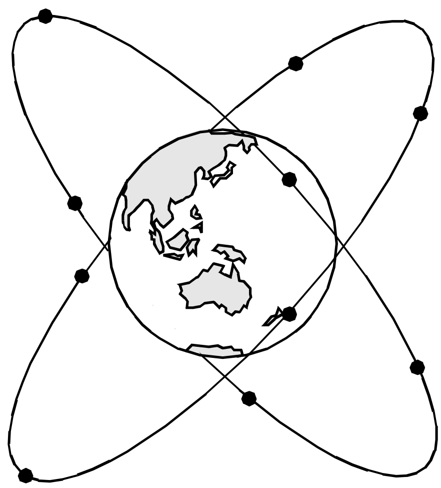
Voice traffic is coded at 4.8 kbit/s and rate $^1/_2$ FEC coding is applied. This 9.6 kbit/s data is spread to 4.833 MHz bandwidth using direct sequence spread spectrum and is binary phase shift key (BPSK) modulated into one of the three sub-bands. CDMA sub-band frequencies are reused in a three-cell frequency reuse pattern. The satellite transponders are all transparent 'bent pipe' transponders routeing traffic between the mobile terminals and an FES. Mobile terminal transmit power is up to 0.5 W or 5 W, depending on the terminal type.

Odyssey is to be promoted as satellite capacity to be sold to terrestrial cellular network operators to 'provide a means by which cellular companies can fill 'holes' in their service areas economically and transparently to the subscriber'. Odyssey space segments will cost US$0.65 per minute to use in addition to terrestrial call routeing charges.

7. ICO

Inmarsat considered becoming partners in other proposals before deciding to design its own system [4] in an announcement of 'Project 21' in September 1991. During 1995 an affiliate company, ICO Global Communications Ltd, London, was spawned to build and operate 'Inmarsat-P'. ICO's final cost is expected to be about US$2.6 billion for the space segment and FESs. US$1.4 billion of investment has already been secured from all over the world, with Asian funding dominating in contrast to the US proposals and Inmarsat's own signatories. While Inmarsat's biggest signatory, Comsat, is also the biggest investor in ICO, the remaining investment is mostly Asian, Middle Eastern and South American, with only a small European content. This reflects the need for cellular extension in these different regions, where Europe will already have a well-developed GSM infrastructure by 2001. Funding is all from service providers such as KDD (Japan), MCN (China), VSNL (India) and CPRM representing operators in Mexico and Brazil.

ICO will launch a ten satellite constellation with five satellites in each of two orbits inclined at 45°, as shown in Fig 7. Extensive beam-forming networks will be used to form 163 beams on each satellite, each of which can be steered to dwell on geographically fixed areas as the satellite moves overhead. ICO will make use of satellite diversity to avoid radio shadowing from terrestrial clutter whenever possible. Figure 8 shows the minimum elevations of two satellites that the constellation guarantees at different latitudes.



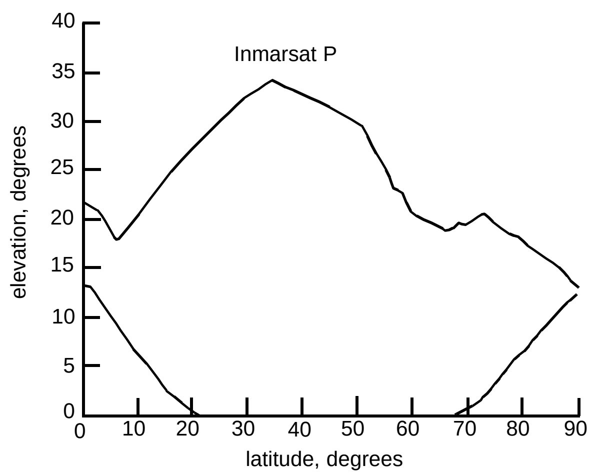


Fig 8 Minimum elevation angles for two ICO satellites.

Mobile uplinks and downlinks will be in the 2 GHz FPLMTS bands, 1.98—2.01 GHz uplinks and 2.17—2.20

Table 1 Mobile satellite proposals.

	Iridium	**Globalstar**	**Odyssey**	**ICO**
Constellation				
Total system cost	\$3.7 bn	\$2.6 bn	\$2.3 bn	\$2.6 bn
Manufacturer	Lockheed	Loral	TRW	Hughes
No of satellites	66	48	12	10
Orbital altitude	780 km	1,414 km	10,354 KM	10,355 km
Orbital inclination	86^o	52^o	55^o	45^o
No of orbital planes	6	8	3	2
Minimum satellite evelation	8^o	0^o (with diversity)	21^o	0^o (with diversity)
Frequencies (GHz)				
Mobile uplinks	1.62 — 1.63	1.61 — 1.63	1.61 — 1.63	1.98 — 2.01
Mobile downlinks	1.62 — 1.63	2.48 — 2.50	2.48 — 2.50	2.17 — 2.20
FES uplinks	19.4 — 19.6	5.09 — 5.25	29.1 — 29.4	6.5
FES downlinks	29.1 — 29.3	6.70 — 7.08	19.3 — 19.6	3.6
Inter-satellite links	23.2 — 23.4	None	None	None
Multiple access				
Scheme	TDMA	CDMA	CDMA	TDMA
Spot beams per satellite	48	16	37	163
Approximate spot beam area	350 000 km^2	2900 000 km ^{2}	2300 000 km^2	950 000 km^2
Channels per beam	23	175	62	28
Channels per million km^2	66	61	27	29
Services				
Voice telephony	4.8 kbit/s	4.8 kbit/s	4.8 kbit/s	4.8 kbit/s
Transparent data (BER $\leq 10^{-3}$)	2.4 kbit/s	7.2 kbit/s	9.6 kbit/s	2.4 kbit/s
Call charge, US\$ per minute	\$3 flat rate	\$0.65 + terrestrial charges	\$0.60 + terrestrial charges	\$1 — \$2
Lead investors				
Service providers	US Sprint Hutchinson Vebacom KMT DDI Japan STET Italy	AirTouch Vodafone Dacom France Telecom	Teleglobe	Inmarsat Comsat KDD Japan MCN China VSNL India CPRM
Manufacturers	Motorola Lockheed	Loral Qualcomm Alcatel Deutsche Aerospace Hyundai	TRW	

GHz downlinks as agreed at the 1995 World Radio Conference. QPSK is used to modulate TDMA carriers allowing 28 mobiles to share each spot beam. The satellites use transparent 'bent-pipe' transponders, routeing traffic from the mobile beams directly to the nearest FES in the same way as Globalstar and Odyssey. Like Iridium, TDMA carrier frequencies are reused on a geographically fixed frequency reuse plan. ICO will use a four-cell reuse pattern. Unlike Iridium, beams can be steered to dwell on cells in the frequency reuse pattern.

8. Conclusions

A summary of the four competing systems is shown in Table 1. Each of the systems offers only a li mited number of channels to cover enormous geographical areas. While global coverage is effectively achieved, use of satellite channels needs to be managed carefully so that customers with no choice but to use satellite can gain access to a satellite channel without being blocked by users who could be using terrestrial cellular networks. This is why close relationships with cellular service providers are being sought and dual-mode handsets that default to cellular wherever possible are being developed.

The result of such partnerships should greatly enhance mobile communications in rural areas all over the world and in areas where communications infrastructure is unavailable because of disaster or economic reasons.

Acknowledgements

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References

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2 Loral Cellular Systems Corp: ‘Application for authority to construct Globalstar’, to the FCC (June 1990 and refiled November 1994).

3 TRW Inc: ‘Application for authority to construct Odyssey’, to the FCC (May 1991 and refiled November 1994).

4 ICO Global Communications Ltd: WorldWide Web pages at http://www.i-co.co.uk/ (1996).

Bob Finean has worked in the aerospace industry since 1985, designing control systems for Rolls Royce RB211 jet engines on Boeing 747 and 767 aircraft. He joined BT Laboratories to pursue his interest in satellite communications after being awarded a Masters degree in electronic and electrical engineering from the University of Leeds in 1990.

His work for BT has used VSAT, Inmarsat and fixed satellite services, including interconnection with a variety of fixed networks using a wide range of protocols and signalling. Since 1991 he has specialised in future mobile satellite systems. He worked at KDD R&D Laboratories in Japan during 1993. This collaborative work was on satellite access to FPLMTS — the third generation mobile systems anticipated around 2005. He has submitted his Doctorate thesis at the University of Surrey, based on his work on FLPMTS and satellites. He is a Chartered Engineer and a Member of the IEE and the IEEE.